SMART STAMPEDE MONITORING AND PREVENTION SYSTEM

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INTRODUCTION

- Stampedes happen during large gatherings, causing injuries and deaths.
- Our project uses smart sensors and technology to detect dangerous crowd behavior.
- Real-time alerts allow quick action to avoid stampedes.
- Monitoring crowd movement can help prevent such disasters.

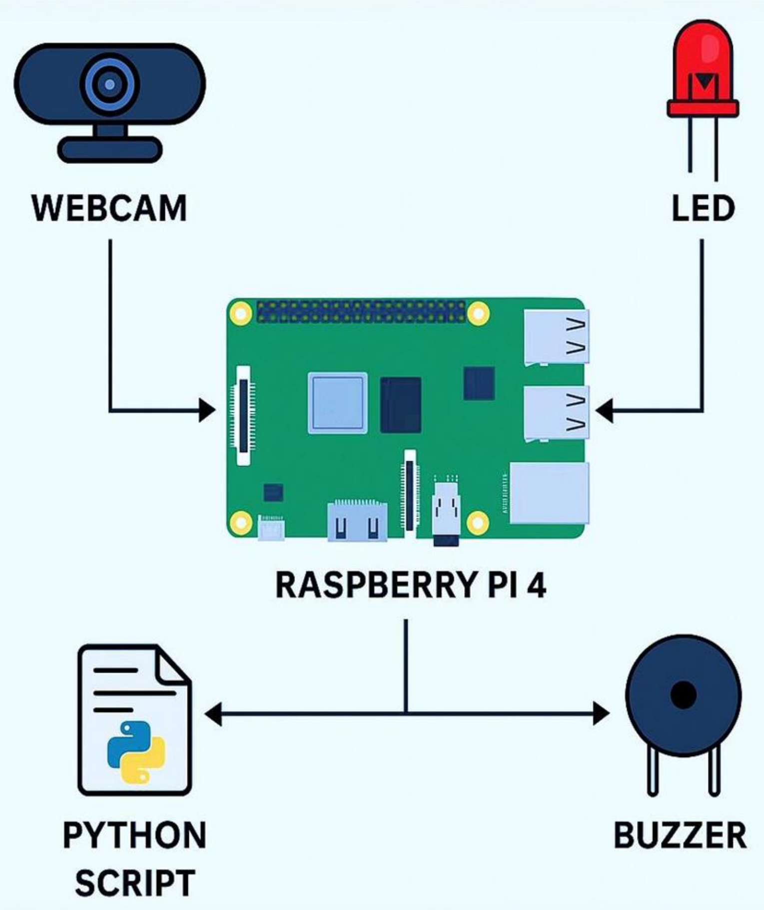
PROBLEM STATEMENT

- Large crowds at events can become dangerous if not monitored.
- Sudden movements or panic can lead to stampedes, causing injuries and deaths.
- There is no effective real-time system to monitor crowd density and prevent stampedes.
- A smart solution is needed to detect risky situations early and alert authorities.

OBJECTIVE

- To design a smart system that monitors crowd density in real-time.
- To detect early signs of stampedes before they happen.
- To send instant alerts to authorities for quick action.
- To ensure the safety of people in large gatherings.

DIAGRAM



COMPONENTS

Hardware :

Raspberry Pi 4

Webcam

LED

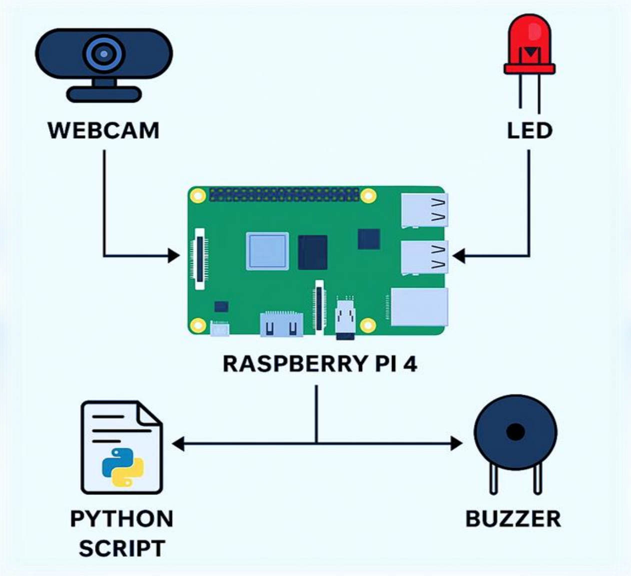
Buzzer

Software :

Python

OpenCV

Raspbian OS



RASPBERRY PI 4

- **Processor:** Quad-Core ARM Cortex-A72
- **RAM :** Available in 2GB, 4GB, and 8GB .
- **Storage:** Uses a MicroSD card for OS and data storage.
- **Connectivity:** Supports Wi-Fi, Bluetooth .
- **GPIO Pins:** 40 pins for connecting sensors, cameras, and other devices.
- **Usage:** Ideal for IoT applications, automation, real-time monitoring, and smart systems.

WEBCAM

- **Model:** Zebronics HD Webcam
- **Resolution:** 720p HD for clear video quality.
- **Built-in Microphone:** Supports audio capture during video streaming.
- **Plug and Play:** Easy USB connectivity, no extra drivers required .
- **Usage:** Ideal for video conferencing, surveillance, and live monitoring in projects.

LED

- **Type:** Small electronic component that emits light when powered.
- **Usage:** Used as an indicator for alerts, status signals, and notifications

  Visual alerts for crowd monitoring, status indicators for system operations.
- **Power Requirement:** Low power consumption, typically 2V–3V.
- **Connectivity:** Easily connected to **GPIO Pins of Raspberry Pi** for control via Python.

BUZZER

Type: Small electronic device that produces sound when powered.

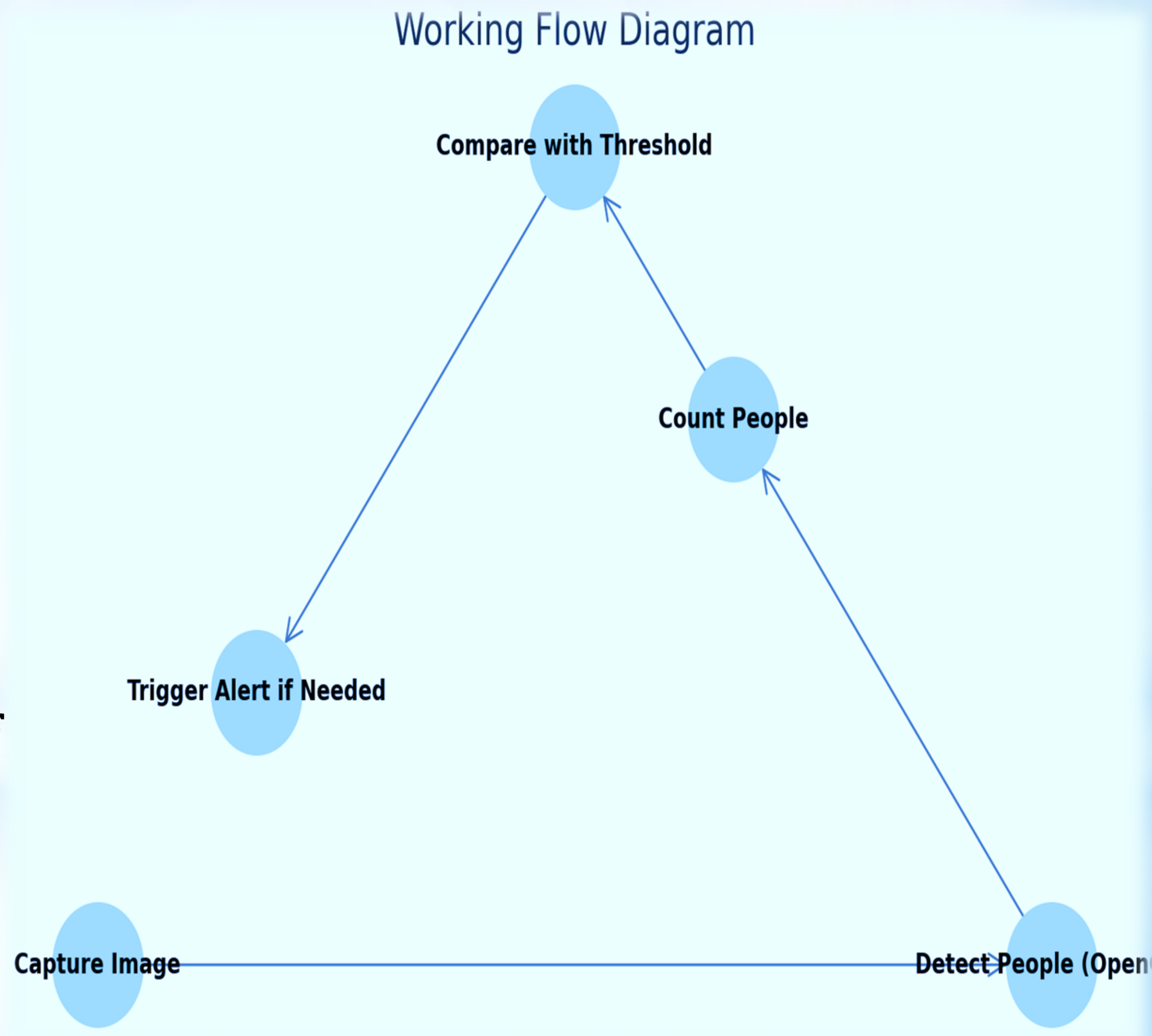
Usage: Used for audio alerts, warnings, and notifications.

Power Requirement: Operates typically at 3V–5V, suitable for Raspberry Pi.

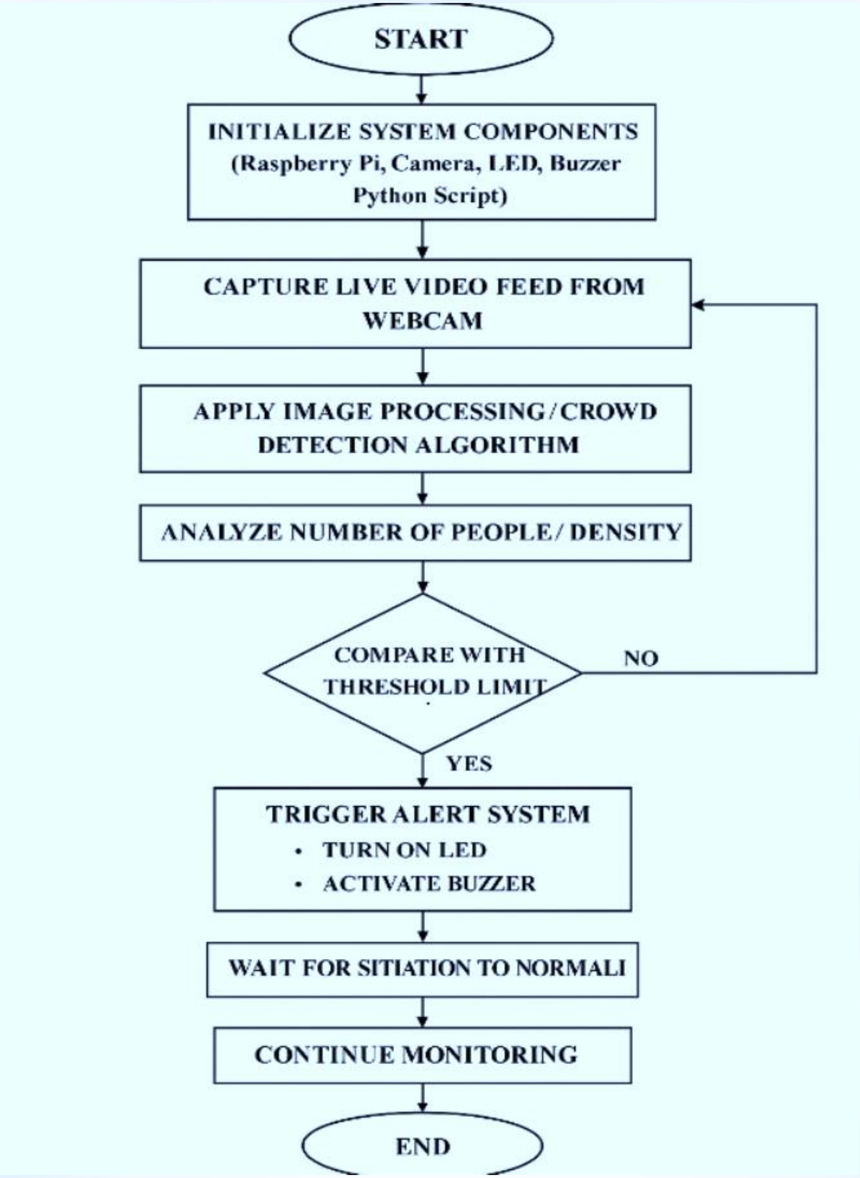
Connectivity: Can be connected to **GPIO Pins of Raspberry Pi** for controlled sound output.

WORKING FLOW

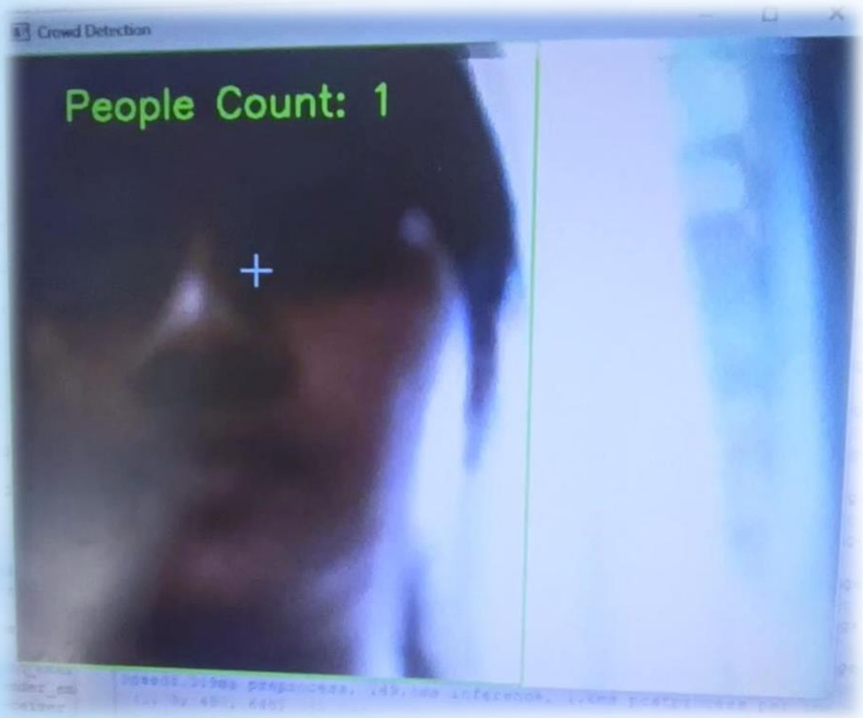
- Detect People (using OpenCV)
- Count People
- Compare with Threshold
- Trigger Alert if Needed
- Capture Image



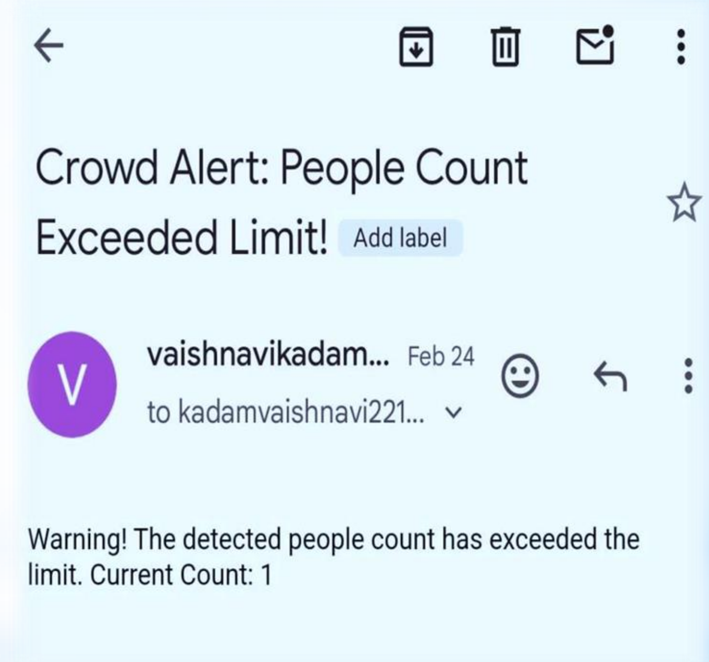
FLOW CHART


START
INITIALIZE SYSTEM COMPONENTS
(Raspberry Pi, Camera, LED, Buzzer
Python Script)
CAPTURE LIVE VIDEO FEED FROM
WEBCAM
APPLY IMAGE PROCESSING/CROWD
DETECTION ALGORITHM
ANALYZE NUMBER OF PEOPLE/ DENSITY
COMPARE WITH
THRESHOLD LIMIT
NO
YES
TRIGGER ALERT SYSTEM
• TURN ON LED
• ACTIVATE BUZZER
WAIT FOR SITIATION TO NORMALI
CONTINUE MONITORING
END

OUTPUT

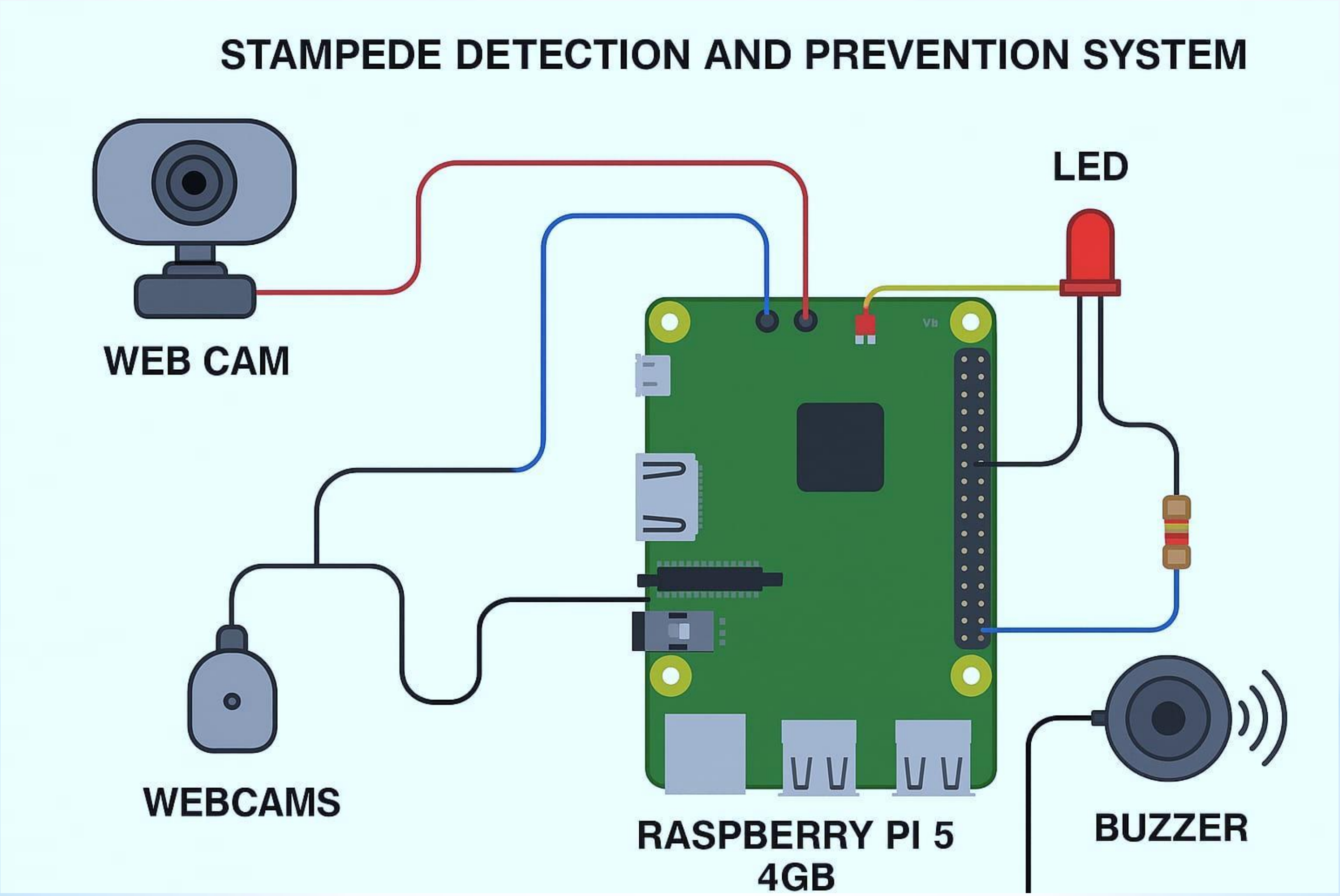


Crowd Detection
People Count: 1



Crowd Alert: People Count Exceeded Limit!
Add label
vaishnavikadam...
Feb 24
V
to kadamvaishnavi221...
Warning! The detected people count has exceeded the limit. Current Count: 1

SYSTEM


STAMPEDE DETECTION AND PREVENTION SYSTEM
LED
WEB CAM
WEBCAMS
RASPBERRY PI 5
4GB
BUZZER

APPLICATIONS

- Temples and Religious Events
- Sports Stadiums
- Concerts and Festivals
- Railway Stations and Airports
- Public Events and Rallies

FUTURE SCOPE

- Real-time aerial monitoring of large crowds.
- Use of Artificial Intelligence to predict crowd behavior.
- Instant alerts and live updates for event organizers and security.
- Real-time location-based crowd analysis
- Store and analyze crowd data for better predictions and planning.

CONCLUSION

This project provides a smart, low-cost solution for real-time crowd monitoring and stampede prevention. It ensures public safety through automation using Raspberry Pi and OpenCV, making it ideal for use in public spaces.

❖ THANK YOU